

Erdős Problem #1093

EP-1093

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

For $n \geq 2k$ we define the deficiency of $\binom{n}{k}$ as follows. If $\binom{n}{k}$ is divisible by a prime $p \leq k$ then the deficiency is undefined. Otherwise, the deficiency is the number of $0 \leq i < k$ such that $n - i$ is k -smooth, that is, divisible only by primes $\leq k$.

Are there infinitely many binomial coefficients with deficiency 1? Are there only finitely many with deficiency > 1 ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>